

Часть 2. Задача 1.

$$1) z = \frac{2}{1+i} - \frac{(1+i)(2-2i)}{(1-i)(1-2i)} = \frac{2}{1+i} - \frac{4}{-1-3i} = \frac{2-2i}{2} - \frac{-4+12i}{10} = 1,4-2,2i.$$

$$2) z = \frac{\sqrt{3}+i}{2-i\sqrt{5}} - \frac{(1+i)(\sqrt{3}+2i)}{4} = \frac{2\sqrt{3}+i\sqrt{15}+2i-\sqrt{5}}{9} - \frac{\sqrt{3}+2i+i\sqrt{3}-2}{4} =$$

$$= \frac{14\sqrt{3}+4i\sqrt{15}+14i-4\sqrt{5}-9\sqrt{3}-18i-9i\sqrt{3}+18}{63} = \frac{5\sqrt{3}-4\sqrt{5}+18}{63} + i \frac{4\sqrt{15}-9\sqrt{3}-4}{63}.$$

$$|z| = \sqrt{a^2+b^2} \quad \operatorname{tg} \varphi = \frac{b}{a}$$

$$3) z = 1+i\sqrt{3} \quad |z|=2 \quad \operatorname{tg} \varphi = \sqrt{3} \rightarrow \varphi = \frac{\pi}{3} \quad z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 2e^{i\frac{\pi}{3}}$$

$$4) z = -1+i\sqrt{3} \quad |z|=2 \quad \operatorname{tg} \varphi = -\sqrt{3} \rightarrow \varphi = -\frac{\pi}{3} \quad z = 2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3})) = 2e^{-i\frac{\pi}{3}}$$

$$5) z = -1-i\sqrt{3} \quad |z|=2 \quad \varphi = -\frac{2\pi}{3} \quad z = 2(\cos(-\frac{2\pi}{3}) + i \sin(-\frac{2\pi}{3})) = 2e^{-i\frac{2\pi}{3}} = 2e^{-i(\frac{2\pi}{3}+2\pi k)}, k \in \mathbb{Z}$$

$$6) z = \sqrt{3}-i \quad |z|=2 \quad \varphi = -\frac{\pi}{6} \quad z = 2(\cos(-\frac{\pi}{6}) + i \sin(-\frac{\pi}{6})) = 2e^{-i\frac{\pi}{6}}$$

$$7) z^4 - zi = 0 \rightarrow z(z^3 - i) = 0 \rightarrow \begin{cases} z=0 \\ z=\sqrt[3]{i} \end{cases} \rightarrow \begin{cases} z=0 \\ z=\sqrt[3]{7}e^{i(\frac{\pi}{6}+\frac{2\pi k}{3})} \end{cases} \rightarrow \begin{cases} z=0 \\ z=e^{i(\frac{\pi}{6}+\frac{2\pi k}{3})} \end{cases}, k=0,1,2$$

$$\text{Ответ: } \{0; e^{i(\frac{\pi}{6}+\frac{2\pi k}{3})} / k=0,1,2\}$$

$$8) z^6 - 8 = 0 \rightarrow z^6 = 8(\cos 0 + i \sin 0) = 8(\cos 0 + i \sin 0) \rightarrow$$

$$\rightarrow \begin{cases} r^6 = 2\sqrt{2} \\ 6\varphi = 0 + 2\pi k \end{cases} \rightarrow \begin{cases} r = \sqrt[6]{2\sqrt{2}} \\ \varphi = \frac{\pi k}{3} \end{cases} \rightarrow z = \sqrt[6]{2\sqrt{2}} e^{i\frac{\pi k}{3}}, k=0,5$$

$$\text{Ответ: } \{\sqrt[6]{2\sqrt{2}} e^{i\frac{\pi k}{3}} / k=0,5\}$$

$$9) z^4 + iz = 0 \rightarrow z(z^3 + i) = 0 \rightarrow \begin{cases} z=0 \\ z=\sqrt[3]{-i} \end{cases} \rightarrow \begin{cases} z=0 \\ z=e^{-i(\frac{\pi}{6}+\frac{2\pi k}{3})} \end{cases}, k=0,1,2.$$

$$\text{Ответ: } \{0; e^{-i(\frac{\pi}{6}+\frac{2\pi k}{3})} / k=0,1,2\}.$$

$$10) z^5 + z = 0 \rightarrow z(z^4 + 1) = 0 \rightarrow \begin{cases} z=0 \\ z^4 = -1 \end{cases} \rightarrow \begin{cases} z=0 \\ z = (\cos(\frac{\pi}{4} + 2\pi k) + i \sin(\frac{\pi}{4} + 2\pi k)) \end{cases} \rightarrow$$

$$\rightarrow \begin{cases} z=0 \\ z = e^{i(\frac{\pi}{4}+2\pi k)}, k=0,3 \end{cases} \quad \text{Ответ: } \{0; e^{i(\frac{\pi}{4}+2\pi k)} / k=0,3\}.$$

Задача 2.

Разложить в ряд Лорана: $\frac{1}{1-q} = \sum_{k=0}^{\infty} q^k, |q| < 1$

$$1) f(z) = \frac{1}{z^2(z-3)} \quad \text{в конуре } 0 < |z| < 3 \quad -1/9.$$

$$\frac{A}{z} + \frac{B}{z-3} \rightarrow A(z-3) + Bz^2 = 1 \rightarrow A = -1/9, B = -1/3, C = -$$

$$f(z) = -\frac{1}{9}z^{-2} - \frac{1}{9}z + \frac{1}{9(z-3)} = -\frac{1}{9}z^{-2} - \frac{1}{9}z - \frac{1}{9} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{9}z^{-2} - \frac{1}{9}z - \frac{1}{9} \sum_{k=0}^{\infty} (\frac{z}{3})^k =$$

$$= -\frac{1}{9}z^{-2} - \frac{1}{9}z - \sum_{k=0}^{\infty} \frac{z^{k+2}}{3^{k+3}} \quad \frac{1}{z-3} = \frac{-1}{3-z} = -\frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}}$$

$$2) f(z) = \frac{1}{z^2(z-3)} \quad \text{в окрестности } z = \infty \rightarrow [0 < |z| < 3]$$

$$f(z) = -\frac{1}{3}z^{-2} - \frac{1}{9}z + \frac{1}{9(z-3)} = -\frac{1}{3}z^{-2} - \frac{1}{9}z + \frac{1}{9z} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{3}z^{-2} - \frac{1}{9}z + \sum_{k=0}^{\infty} \frac{3^{k-2}}{z^{k+1}} = \sum_{k=2}^{\infty} \frac{3^{k-2}}{z^{k+1}}.$$

$$f(z) = -\frac{1}{3}z^{-2} - \frac{1}{9}z + \sum_{k=0}^{\infty} \frac{3^{k-2}}{z^{k+1}} = -\frac{1}{3}z^{-2} - \frac{1}{9}z + \frac{1}{9}z + \frac{1}{3}z^2 + \dots = \sum_{k=2}^{\infty} \frac{3^{k-2}}{z^{k+1}}.$$

- 3) $f(z) = \frac{z+1}{z^2(z-1)}$ в кольце $0 < |z| < 1$
 $f(z) = \frac{\frac{z}{z^2}}{z^2(z-1)} + \frac{1}{z^2(z-1)} = \frac{1}{z^2(z-1)} + \frac{1}{z^2(z-1)} = \frac{1}{z^2-1} - \frac{1}{z} - \frac{1}{z} - \frac{1}{z^2} + \frac{1}{z-1} = \frac{-z}{z^2} - \frac{1}{z^2} - \frac{z}{1-z} = \frac{-1}{z^2} - \frac{z}{1-z} = \frac{-1}{z^2} - \sum_{k=0}^{\infty} z^k$
- 4) $f(z) = \frac{z+1}{z^2(z-1)}$ в окрестности $z = \infty$
 $f(z) = \frac{-z}{z^2} - \frac{1}{z} + \frac{z}{z-1} = \frac{-z}{z^2} - \frac{1}{z} + \frac{z}{z} \cdot \frac{1}{1-1/z} = \frac{-z}{z^2} - \frac{1}{z} + \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} = \frac{1}{z^2} + \sum_{k=3}^{\infty} \frac{1}{z^k}$
- 5) $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ в кольце $0 < |z-1| < 3$
 $\frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{z+2} \rightarrow A(z^2+z-2) + B(z+2) + C(z-1)^2 = 4z \rightarrow A = -8/9, B = 4/3, C = -8/9$
 $f(z) = \frac{8/9}{z-1} + \frac{4/3}{(z-1)^2} - \frac{8/9}{z+2} = \frac{8/9}{z-1} + \frac{4/3}{(z-1)^2} - \frac{8}{9} \cdot \frac{1}{1+(z-1)/3} = \frac{8/9}{z-1} + \frac{4/3}{(z-1)^2} - \frac{8}{9} \sum_{k=0}^{\infty} \left(-\frac{z-1}{3}\right)^k$
- 6) $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ в окрестности $z = \infty$
 $f(z) = \frac{8/9}{z-1} + \frac{4/3}{(z-1)^2} - \frac{8/9}{z+2} = \frac{8}{9} \cdot \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} - \frac{4}{3} \sum_{k=0}^{\infty} \frac{-k-1}{z^{k+2}} - \frac{8}{9} \sum_{k=0}^{\infty} \frac{z^k(-1)^k}{z^{k+1}}$
 $I: \frac{8/9}{z-1} = \frac{8}{9} \cdot \frac{1}{z(1-1/z)} = \frac{8}{9} \cdot \frac{1}{z} \cdot \sum_{k=0}^{\infty} \frac{1}{z^k} = \frac{8}{9} \cdot \sum_{k=0}^{\infty} \frac{1}{z^{k+1}}$
 $II: \frac{4/3}{(z-1)^2} = -\frac{4}{3} \left(\frac{1}{z-1}\right)' = -\frac{4}{3} \left(\sum_{k=0}^{\infty} \frac{1}{z^{k+1}}\right)' = -\frac{4}{3} \sum_{k=0}^{\infty} \frac{-k-1}{z^{k+2}}$
 $III: -\frac{8/9}{z+2} = -\frac{8}{9} \cdot \frac{1}{z(1+2/z)} = -\frac{8}{9} \sum_{k=0}^{\infty} \frac{z^k(-1)^k}{z^{k+1}}$
- 7) $f(z) = \frac{z^2}{(z-1)(z+2)}$ в кольце $1 < |z| < 2$
 $f(z) = \frac{2/3}{z-1} + \frac{4/3}{z+2} = \frac{2}{3} \cdot \frac{1}{z} \cdot \frac{1}{1-1/z} + \frac{4}{3} \cdot \frac{1}{z} \cdot \frac{1}{1-(2/z)} = \frac{2}{3} \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} + \frac{4}{3} \sum_{k=0}^{\infty} \frac{2^k(-1)^k}{z^{k+1}}$
- 8) $f(z) = \frac{z^2}{(z-1)(z+2)}$ в окрестности $z = \infty$
 $f(z) = \frac{2/3}{z-1} + \frac{4/3}{z+2} = \frac{2}{3} \cdot \frac{1}{z} \cdot \frac{1}{1-1/z} + \frac{4}{3} \cdot \frac{1}{z} \cdot \frac{1}{1-(2/z)} = \frac{2}{3} \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} + \frac{4}{3} \sum_{k=0}^{\infty} \frac{2^k(-1)^k}{z^{k+1}}$
- 9) $f(z) = \frac{1}{(z^2+1)^2} + \sin \frac{1}{z} + 6z^5$ в окрестности $z = \infty$
 $\frac{1}{(z^2+1)^2} = -\frac{1}{z^2} \cdot \left(\frac{1}{z^2+1}\right)' = -\frac{1}{z^2} \left(\frac{1}{z^2(1+1/z^2)}\right)' = -\frac{1}{z^2} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{z^{2k+2}}\right)' = -\frac{1}{z^2} \left(\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 2k}{z^{2k+1}}\right) = -\frac{1}{z^2} \sum_{k=1}^{\infty} \frac{(-1)^{k+1} 2k}{z^{2k+1}} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{z^{2k+3}}$
 $\sin \left(\frac{1}{z}\right) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{-(2k+1)}$
 $f(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{-(2k+1)} - \sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{z^{2k+3}} + 6z^5$
- 10) $f(z) = \frac{1}{(z^2+1)^2} + \sin \frac{1}{z} + 6z^5$ в окрестности $z = 0$
 $\frac{1}{(z^2+1)^2} = -\frac{1}{z^2} \cdot \left(\frac{1}{z^2+1}\right)' = -\frac{1}{z^2} \left(\sum_{k=0}^{\infty} (-1)^k z^{2k}\right)' = -\frac{1}{z^2} \sum_{k=1}^{\infty} (-1)^k 2k z^{2k-1} = \sum_{k=1}^{\infty} (-1)^{k+1} z^{2k-3} \cdot k$
 $\sin \left(\frac{1}{z}\right) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{-(2k+1)}$
 $f(z) = \sum_{k=1}^{\infty} (-1)^{k+1} z^{2k-3} \cdot k + \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{-(2k+1)} + 6z^5$

Задание 3.

Вычислить интеграл по границе области D с пом. вычетов:
 Используем основную теорему о вычетах: $\oint_D f(z) dz = 2\pi i \cdot \sum_{z_k \in D} \text{res}_{z_k} f(z)$, $z_k \in D$,
 где $\text{res}_{z_k} f(z)$ — вычет ф-ции $f(z)$ в особой т. $z = z_k$.

- 1) $\int_D \frac{dz}{z^2(z-3)}$, $D: |z| < 4$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty$ — устр.
 $\int_D \frac{dz}{z^2(z-3)} = 2\pi i \cdot \sum_{z_k \in D} \text{res}_{z_k} f(z) = -2\pi i \cdot \text{res}_{z=3} f(z) = -2\pi i \cdot \lim_{z \rightarrow 3} z(f(\infty) - f(z)) = -2\pi i \cdot \lim_{z \rightarrow 3} z(0 - f(z)) =$
 $= 2\pi i \cdot \lim_{z \rightarrow 3} z f(z) = 2\pi i \cdot \lim_{z \rightarrow 3} \frac{z}{z^2(z-3)} = 2\pi i \cdot \lim_{z \rightarrow 3} \frac{1/z^2}{1-3/z} = 2\pi i \cdot 0 = 0$
- 2) $\int_D \frac{\sin \sqrt{z}}{1+z} dz$, $D: |z| < 2$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty$ — устр.
 $\int_D f(z) dz = 2\pi i \cdot \sum_{z_k \in D} \text{res}_{z_k} f(z) = -2\pi i \cdot \text{res}_{z=-1} f(z) = -2\pi i \cdot \lim_{z \rightarrow -1} z(f(\infty) - f(z)) = -2\pi i \cdot \lim_{z \rightarrow -1} z(0 - f(z)) =$
 $= 2\pi i \cdot \lim_{z \rightarrow -1} \frac{z \sin \sqrt{z}}{1+z} = 2\pi i \cdot \lim_{k \rightarrow 0} \frac{2/k \cdot \sin k}{1+1/k} = 2\pi i \cdot \lim_{k \rightarrow 0} \frac{1/k \cdot k \cdot \sin k}{(1+1/k) \cdot k} = 2\pi i \cdot \lim_{k \rightarrow 0} \frac{k}{(k+1)/k} = 2\pi i \cdot \lim_{k \rightarrow 0} \frac{-dk}{k+1} = 0$
- 3) $\int_D \frac{z+1}{z^2(z-1)} dz$, $D: |z| < 2$
 $\int_D \frac{z+1}{z^2(z-1)} dz = 2\pi i \cdot \sum_{z_k \in D} \text{res}_{z_k} f(z) = -2\pi i \cdot \text{res}_{z=1} f(z) = -2\pi i \cdot \lim_{z \rightarrow 1} z(0 - f(z)) = 2\pi i \cdot \lim_{z \rightarrow 1} \frac{z+1}{z(z-1)} = 0$

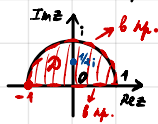
- 4) $\int_{\gamma} z \sinh \frac{z+1}{z-1} dz$, $D: |z| < 2$
 $f(z) = z \sinh \frac{z+1}{z-1} = \left| \frac{w=z-1}{w+1} \right| = (w+1) \sinh \frac{w+2}{w} = (w+1) \sinh \left(1 + \frac{2}{w} \right) = (w+1) \sinh 1 \cosh \frac{2}{w} + (w+1) \cosh 1 \sinh \frac{2}{w} = \tilde{f}(w)$
 $\tilde{f}(w) = \sinh 1 (w+1) \left(1 - \frac{1}{2} \frac{2^2}{w^2} + \dots \right) + \cosh 1 (w+1) \left(\frac{2}{w} + \frac{1}{6} \frac{2^3}{w^3} + \dots \right)$
 $C_{-1} = 2 \cos 1 - 2 \sin 1 = \operatorname{res} f(z) \Rightarrow \int_{\gamma} f(z) dz = 2\pi i (2 \cos 1 - 2 \sin 1) = 4\pi i (\cos 1 - \sin 1)$
- 5) $\int_{\gamma} \frac{4z = f(z)}{(z-1)^2(z+2)} dz$, $D: |z| < 3$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{4z}{(z-1)^2(z+2)} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{4z^2}{(z-1)^2(z+2)} = 0$
- 6) $\int_{\gamma} \frac{z^2 = f(z)}{z^2-1} dz$, $D: |z| < 2$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{z^2}{z^2-1} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{z^2}{z^2-1} = 2\pi i$
- 7) $\int_{\gamma} \frac{dz = f(z)}{(z-1)^2(z+2)} dz$, $D: |z| < 3$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{dz}{(z-1)^2(z+2)} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{dz}{(z-1)^2(z+2)} = 4\pi i$
- 8) $\int_{\gamma} \frac{z = f(z)}{(z-1)^2(z^2-2)} dz$, $D: |z| < 2$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{z}{(z-1)^2(z^2-2)} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{z^2}{(z-1)^2(z^2-2)} = 0$
- 9) $\int_{\gamma} \frac{z = f(z)}{(z^2+2)(z-2)} dz$, $D: |z| < 3$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{z}{(z^2+2)(z-2)} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{z^2}{(z^2+2)(z-2)} = 0$
- 10) $\int_{\gamma} \frac{z = f(z)}{(z^2+1)(z+2)} dz$, $D: |z| < 3$ $\lim_{z \rightarrow \infty} f(z) = 0 \Rightarrow z = \infty - \text{ycmp.}$
 $\int_{\gamma} f(z) dz = -2\pi i \operatorname{res} f(z) = -2\pi i \lim_{z \rightarrow \infty} z \left(0 - \frac{z}{(z^2+1)(z+2)} \right) = 2\pi i \lim_{z \rightarrow \infty} \frac{z^2}{(z^2+1)(z+2)} = 0$

Задача 4.

Найти образ области при данном отображении:

- 1) $D = \{0 < \operatorname{Re} z < 1\}$, $w = \frac{z-1}{z+2}$
 $\omega_1: w_1 = w(z_1) = w(0) = -\frac{1}{2}$
 $\omega_2: w_2 = w(z_2) = w(1) = 0$
 $w_3 = w(z_3) = w(i\infty) = 1$
 $w_4 = w(z_4) = w(-i) = -\frac{1}{5} - \frac{3}{5}i$
 $w_5 = w(z_5) = w(-1-i) = \frac{1}{10} - \frac{3}{10}i$
 $w_6 = w(z_6) = w(i\infty) = 1$
- 2) $D = \{z-1 < 2, \operatorname{Im} z < 0\}$, $w = \frac{z+1}{z-3}$
 $\omega_1: w_1 = w(1) = -1$
 $\omega_2: w_2 = w(-1) = 0$
 $\omega_3: w_3 = w(1-2i) = i$
 $\omega_4: w_4 = w(-1-i) = 0$
- 3) $D = \{z-1 > 2, \operatorname{Im} z < 0\}$, $w = \frac{1}{z}$
 $\omega_1: w_1 = w(-\sqrt{3}) = -\frac{1}{\sqrt{3}}$
 $\omega_2: w_2 = w(\sqrt{3}) = \frac{1}{\sqrt{3}}$
 $\omega_3: w_3 = w(-i) = i$
 $\omega_4: w_4 = w(3i) = -\frac{1}{3}i$
- 4) $D = \{|z| < 1, \operatorname{Re} z > 0\}$, $w = \frac{1-z}{z+2}$
 $\omega_1: w_1 = w(0) = \frac{1}{2}$
 $\omega_2: w_2 = w(-1) = 2$
 $\omega_3: w_3 = w(i) = \frac{1}{5} - \frac{3}{5}i$
 $\omega_4: w_4 = w(-i) = \frac{1}{5} + \frac{3}{5}i$

5) $D = \{z | z < 1, \operatorname{Im} z > 0\}, w = \frac{1-z}{1+z}$



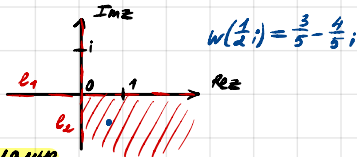
$l_1: w_1 = w(0) = 1$

$l_2: w_1 = w(1) = 0$

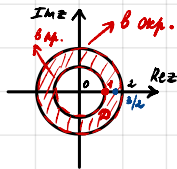
$w_2 = w(1) = 0$

$w_2 = w(i) = -i$

Границы прех. ч. особую т. $\Rightarrow$ отобр. в прямые.



6) $D = \{z | 1 < |z| < 2\}, w = \frac{z}{z-1}$

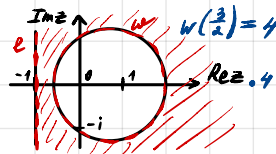


$l_1: w_1 = w(-1) = -1$

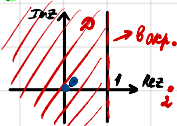
$l_2: w_1 = w(-2) = -\frac{2}{3}$

$w_2 = w(i) = -1-i$

$w_2 = w(2) = 2$

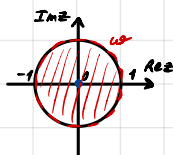


7) $D = \{z | \operatorname{Re} z < 1\}, w = \frac{z}{z-2}$



$l_1: w_1 = w(1) = -1$

$w_2 = w(i\infty) = 1$



$w(0) = 0$

8) $D = \{z | |z-1| < 2\}, w = \frac{z+1}{z-2}$

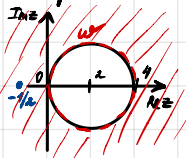
$z=2$ - особая точка

Граница $|z-1|=2$ не прех. ч. особую т. $\Rightarrow$ она отобр. в окр..

$l_1: w_1 = w(-1) = 0$
граница. прех. ч.

$w_2 = w(3) = 1$

(находим модуль, чтобы кот. модуль отобр. окр.-т.)



$z_0 = 0$ - произв. м.

$w(z_0) = w(0) = -1/2 \Rightarrow$

$\Rightarrow$ найти область

9) $D = \{z | 1 < \operatorname{Re} z < 3, 0 < \operatorname{Im} z < \pi/2\}, w = e^z$

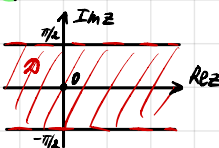


$z = x + iy, x \in (1, 3), y \in (0, \pi/2) \rightarrow$

$\rightarrow |z| = e^x \in (e, e^3), \varphi \in (0, \pi/2)$



10) $D = \{z | |\operatorname{Im} z| < \pi/2\}, w = e^z$



$z = x + iy, x \in (-\infty, +\infty), y \in (-\pi/2, \pi/2) \rightarrow$

$\rightarrow |z| = e^z \in (0, +\infty), y \in (-\pi/2, \pi/2)$

